

## What is a Bash Shell?

**Bash (Bourne Again SHell)** is a command-line interpreter used in Linux and Unix systems. It allows users to execute commands, write scripts, automate tasks, and manage system operations efficiently.

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## Why is echo Used in Bash Shell?

The echo command is used to display output on the terminal. It is commonly used to:

- Print messages
- Display variable values
- Show script execution status
- Debug scripts

Example:

```
echo "Hello World"
```

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## Bash Script to Display Username as Lipsika

```
#!/bin/bash
```

```
echo "Username is Lipsika"
```

Explanation:

- `#!/bin/bash` specifies the script interpreter
  - `echo` prints the required output on the screen
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## Compare Variables and Environment Variables

| Variables                            | Environment Variables                     |
|--------------------------------------|-------------------------------------------|
| Defined within a script or shell     | Defined at system or user level           |
| Accessible only in the current shell | Accessible across child processes         |
| Temporary in nature                  | Persist for the session                   |
| Example: <code>name=Lipsika</code>   | Example: <code>export name=Lipsika</code> |

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**Summary:**

**Bash shell enables automation through scripting, echo helps display output, and understanding variables vs environment variables is essential for effective script execution.**